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Representation of the influence of soil structure on hydraulic conductivity prediction

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ABSTRACT

The significant impact of soil structure on soil hydraulic properties and then on the associated water and solute transport is well recognized. However, existing soil hydraulic models that account for the effect of soil structure often at the cost of overparameterization, preventing further application in simulation of water and solute transport on large scales. In this study, we developed a new model that considers the effects of soil structure near saturation in prediction of hydraulic conductivity while introducing no new free parameters. Testing with 52 soil samples that include different soil types showed that the new model considerably improved the prediction of hydraulic conductivity compared to an earlier version of the model that did not take into account the effects of soil structure. The reported root-mean-square error (RMSE) of the new model is 0.78 cm d^{-1}, much lower than the 0.90 cm d^{-1} when the earlier version of model is used. When the new model was used for fitting observations, it performed very well, with an RMSE of only 0.23 cm d^{-1} remained for 52 soil samples. This new soil hydraulic model provides a simple and practical way to incorporate the influence of soil structure near saturation in water and solute transport simulations.

1. Introduction

Soil structure, which refers to the arrangement of soil pore space (e.g., Rabot et al., 2018; Meurer et al., 2020), is an important property that is generally related to the effects of biological activity, abiotic factors, or tillage practices. The significant influence of soil structure, such as the presence of macropores, on soil hydraulic functions, including the soil water retention curve (SWRC) and the hydraulic conductivity curve (HCC), and therefore on the associated water and solute transport has long been recognized (e.g., Dexter, 1988; Smettem et al., 1991; Gerke & van Genuchten, 1993; 1996; Zhang & van Genuchten, 1994; Durner, 1994; Mohanty et al., 1997; Nimmo, 1997; Šimůnek et al., 2003; Dexter et al., 2008; Jarvis, 2007; Beven and Germann, 1982; 2013; Jarvis et al., 2016; Vereecken et al., 2016; Robinson et al., 2019; Nimmo et al., 2021). The significant impact of soil structure on the water cycle has also been recognized for large-scale applications. For example, a recent study by Fatichi et al. (2020) showed that accounting for soil structure in land surface models can considerably alter the infiltration-runoff process, especially in wet and vegetated regions. Bonetti et al. (2021) also demonstrated the crucial role of soil structure in the infiltration-runoff process.

In order to capture the effects of soil structure on water and solute transport, the first task is to develop a soil hydraulic model that takes into account the effects of soil structure. The mostly applied soil hydraulic models, such as the van Genuchten (1980)-Mualem (1976) model, are the unimodal models and do not consider the effects of soil structure. Early studies, such as van Genuchten and Nielsen (1985), Schaap and Leij (2000) and Schaap et al. (2001), have found that when applying the saturated hydraulic conductivity K_s as a matching point in predicting the HCC, a standard method used by the hydrological community, the unimodal models tend to considerably overestimate the unsaturated conductivity, especially for fine-textured soils. van Genuchten and Nielsen (1985) argued that a new matching point below saturation rather than K_s should be applied in predicting the HCC, which unsurprisingly, will underestimate the hydraulic conductivity at and near saturation.

From a hydrologic point of view, the primary effect of soil structure is more pronounced at and near saturated conditions, mostly reflected in the saturated or near saturated hydraulic conductivity. In contrast, soil texture/matrix mainly controls the unsaturated flow. Therefore, a simple way to represent the effects of soil structure in a soil hydraulic model is to separate the hydraulic conductivity function into a soil structure-dominated zone and a soil matrix-dominated zone. Such models developed by Jarvis et al. (1991), Larsbo et al. (2005), Børgesen et al. (2006), and Schaap and van Genuchten (2006) introduced a new parameter of the saturated matrix hydraulic conductivity, above which the soil

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structure controls the water flow. To predict the HCC from SWRC, therefore, two matching points of K_s and saturated matrix hydraulic conductivity are needed. These hydraulic functions were generally used in dual-permeability models that do not require knowledge of the water retention curve near saturation (Larsbo et al., 2005). In contrast to models that focus only on the effects of soil structure on the HCC, bimodal or multimodal soil hydraulic models that were well established back to the early 1990s (e.g., Othmer et al., 1991, Ross & Smettem, 1993; Durner, 1994) capture the effects of soil structure in both the SWRC and HCC functions. Similar to unimodal models, only a matching point of K_s is needed in predicting the HCC in addition to the parameters defined by the SWRC. However, the developed bi-modal models introduce a large number of parameters, generally, requiring at least three more parameters compared to the commonly applied unimodal models. The determination of parameters and then the prediction of HCC rely heavily on the detailed measurements of soil water retention properties, in particular, near saturation, which, however, are often not available. A recent study by Zhang et al. (2022) also showed that fitting solely the SWRC and applying K_s as a matching point yield poor estimation of the HCC when using the bimodal model developed by Durner (1994). They attributed the poor performance to the fact that most information about bimodality is represented in the HCC rather than the SWRC. Additionally, few existing pedotransfer functions (PTFs) that relate soil hydraulic parameters to more easily measured soil information (e.g., soil texture properties) have been developed for bimodal soil hydraulic models that consider the effects of soil structure (e.g., Jarvis et al., 2013; Fatichi et al., 2020). Consequently, the bimodal hydraulic models are rarely used in large-scale applications (Fatichi et al., 2020).

Wang et al. (2022a) recently proposed a new HCC model through building a physical relationship between the soil matrix-controlled conductivity and the SWRC. The new HCC model has the ability to predict the HCC completely from the SWRC without the knowledge of K_s. However, due to not consider the effects of soil structure, the HCC model in Wang et al. (2022a) tends to underestimate the hydraulic conductivity near saturation for most soil samples. Consequently, this study aimed to develop and evaluate a novel soil hydraulic model that considers the influence of soil structure near saturation based on the HCC model developed in Wang et al. (2022a). Specifically, we focus mainly on the impact of soil structure near saturation on the prediction of HCC, although it is evident that soil structure also impacts SWRC. The reason for this is that water flow in macropores is often assumed to be driven mainly by gravity (Gerke, 2006), and the hydraulic conductivity is of greater concern. The new model developed has three attractive properties: (1) it does not introduce additional free parameters compared to the unimodal soil hydraulic models; (2) it has the ability to predict the HCC from the SWRC with the known matching point of K_s, as did the unimodal model and (3) it takes into account the effects of adsorption forces in the dry range.

2. Model development

In this section, we first recall the FXW-M2 model developed in Wang et al. (2022a). Following this, we demonstrate the development of the new model that accounts for the effects of soil structure. The FXW-M2 may underestimate the hydraulic conductivity near saturation due to either the uncaptured impact of soil structure or the underestimated hydraulic conductivity that accounts for the effects of adsorption forces, which is applied as a matching point in estimating the HCC. Therefore, to demonstrate the influence of soil structure, we also introduce the modified van Genuchten (1980)-Mualem (1976) model developed by Ippisch et al. (2006), termed as the MVGM model hereafter. For a given soil sample, when the MVGM model overestimates and the FXW-M2 model underestimates the hydraulic conductivity near saturation, we conclude that the effects of soil structure are important.

2.1. The FXW-M2 model developed by Wang et al. (2022a)

The FXW-M2 model developed by Wang et al. (2022a) is a modification of the original Fredlund and Xing (1994)-Wang et al. (2018) model. The FXW-M2 model introduces a new matching point rather than K_s in estimating the HCC. This new matching point that accounts for the effects of adsorption forces can be estimated directly from the SWRC, therefore, enabling a direct prediction of HCC fully from SWRC.

The SWRC of the FXW-M2 model is written as:

$$S(h) = \frac{\theta}{\theta_s} = \begin{cases} \left[1 - \dfrac{\ln(1 + (h - h_s)/h_r)}{\ln(1 + (h_0 - h_s)/h_r)}\right] \dfrac{\Gamma(h)}{\Gamma(h_s)} & h < h_s \\ 1 & h \geqslant h_s \end{cases} \tag{1}$$

with $\Gamma(h_s)$ being:

$$\Gamma(h_s) = \left(\ln\left(e + |\alpha h_s|^n\right)\right)^{-m} \tag{2}$$

In Equations (1) and (2), θ ($L^3\ L^{-3}$) is the volumetric water content and θ_s ($L^3\ L^{-3}$) is the saturated water content; h (L) is the matric potential; h_r is a shape parameter and is set to -1.5×10^3cm, following Fredlund and Xing (1994); h_0, which is set as -6.3×10^6cm, according to Schneider and Goss (2012), is the matric potential corresponding to zero water content; h_s, set to -1 cm according to Wang et al. (2022a), is introduced to overcome the unrealistic decrease near saturation for fine-textured soils; and α (L^{-1}), n, and m are the fitted parameters.

The HCC of the FXW-M2 model is expressed as:

$$K(h) = \begin{cases} \theta_s b(h_m)\Gamma(h_m)\left(\dfrac{\Gamma(h) - \Gamma(h_0)}{\Gamma(h_m) - \Gamma(h_0)}\right)^l \left[\dfrac{1 - \left(1 - \Gamma(h)^{1/m}\right)^{1-1/n}}{1 - \left(1 - \Gamma(h_m)^{1/m}\right)^{1-1/n}}\right]^2 & h < h_s \\ \theta_s b(h_m)\Gamma(h_m)\left(\dfrac{\Gamma(h_s) - \Gamma(h_0)}{\Gamma(h_m) - \Gamma(h_0)}\right)^l \left[\dfrac{1 - \left(1 - \Gamma(h_s)^{1/m}\right)^{1-1/n}}{1 - \left(1 - \Gamma(h_m)^{1/m}\right)^{1-1/n}}\right]^2 & h \geqslant h_s \end{cases} \tag{3}$$

where l has a default value of 3.5 following Wang et al. (2018); h_m, set to -1.0×10^5cm according to Wang et al. (2022a), is a typical matric potential where soil water is assumed to be held solely by the van der Waals forces; $b(h_m)$, set to 2.693×10^{-6} cm d^{-1}, is a combination of the film thickness and the specific surface area estimation (Wang et al., 2022a). The detailed derivation of $b(h_m)$ can be found in Wang et al. (2022a). Equation (3) indicates that the HCC can be fully predicted from the SWRC, since l and $b(h_m)$ are constants while the other parameters required can be estimated based on Equation (1).

2.2. Accounting for the influence of soil structure—The FXW-M3 model and the FXW-M3-Opt model

Although the FXW-M2 model matches the observations well in the medium to dry water content range, it underestimates the conductivity near saturation for many soil samples (Wang et al., 2022a). This underestimation is attributed mainly to the inability of the FXW-M2 model to account for the effects of macroporosity.

To represent the influence of soil structure in the soil hydraulic model, the first step is to define a critical matric potential of h_a to distinguish the impact from soil structure and soil texture. The corresponding hydraulic conductivity $K(h_a)$, termed also as the saturated matrix hydraulic conductivity, can be estimated by Equation (3), written as

$$K(h_a) = \min\left\{\theta_s b(h_m)\Gamma(h_m)\left(\frac{\Gamma(h_a)-\Gamma(h_0)}{\Gamma(h_m)-\Gamma(h_0)}\right)^l\left[\frac{1-\left(1-\Gamma(h_a)^{1/m}\right)^{1-1/n}}{1-\left(1-\Gamma(h_m)^{1/m}\right)^{1-1/n}}\right]^2, K_s\right\} \quad (4)$$

The maximum limitation of K_s introduced in Equation (4) is to avoid the unrealistic estimation of $K(h_a)$ value by Equation (3). For matric potentials lower than h_a, the hydraulic conductivity can still be estimated by Equation (3). When the matric potential is higher than h_a, the impact of the soil structure becomes to be important. Here, we simply assume that the hydraulic conductivity for a potential higher than h_a can be described as a power function of the saturation degree S following Campbell (1974). When the matric potential is higher than h_a, the conductivity is written as

$$K(h) = K_s S(h)^{n^*} \quad (5)$$

where n^* is a scaling factor.

By equaling Equation (5) to $K(h_a)$ gives the expression of the scaling factor n^*, written as

$$n^* = \frac{\ln(K(h_a)/K_s)}{\ln(S(h_a))} \quad (6)$$

Resubstituting Equation (6) into Equation (5) gives the conductivity for matric potentials higher than h_a, which is written as

$$K(h) = K_s S(h)^{\frac{\ln(K(h_a)/K_s)}{\ln(S(h_a))}} \approx K_s\Gamma(h)^{\frac{\ln(K(h_a)/K_s)}{\ln(\Gamma(h_a))}} \quad (7)$$

Notably, for matric potential that varies from h_a to 0, the saturation degree $S(h)$ is almost the same as $\Gamma(h)$ (Wang et al., 2016). Together with the conductivity described by Equation (3) for a matric potential less than h_a, the new HCC can thus be expressed as

$$K(h) = \begin{cases} K(h_a)\left(\frac{\Gamma(h)-\Gamma(h_0)}{\Gamma(h_a)-\Gamma(h_0)}\right)^l\left[\frac{1-\left(1-\Gamma(h)^{1/m}\right)^{1-1/n}}{1-\left(1-\Gamma(h_a)^{1/m}\right)^{1-1/n}}\right]^2 & h < h_a \\ K_s\Gamma(h)^{\frac{\ln(K(h_a)/K_s)}{\ln(\Gamma(h_a))}} & h \geqslant h_a \end{cases} \quad (8)$$

The value of the threshold potential h_a depends on the diameter of the smallest macropore of the individual soil. The reported h_a varies from approximately −4 cm in Børgesen et al. (2006) to −40 cm in Schaap and van Genuchten (2006). In this study, the exact value of h_a is determined by optimizing the HCC with observations for each soil type. When h_a is determined, the HCC can be directly predicted from the SWRC with known K_s.

For the SWRC, Equation (1) can be applied. Alternatively, we can apply the SWRC model developed originally by Fredlund and Xing (1994). That is, the new model does not require the introduction of h_s as in the FXW-M2 model to overcome the unrealistic decrease in HCC for fine-textured soils (Wang et al., 2022a). The reason for this is that for matric potentials higher than h_a, a new conductivity function that has a fixed lower boundary of $K(h_a)$ is introduced in the new model (Equation 8). As a result, the HCC of the new model no longer shows an abrupt decrease for values of n close to 1 (Fig. 1). Consequently, we can apply the simple SWRC model developed originally by Fredlund and Xing (1994) for the new model, which is written as

$$\theta = \theta_s\left[1-\frac{\ln(1+h/h_r)}{\ln(1+h_0/h_r)}\right]\Gamma(h) \quad (9)$$

Equations (8) and (9) provide a simple way to describe the soil hydraulic properties from saturation to oven dryness. We refer the new model as the FXW-M3 model hereafter. Both the impact of soil structure and soil texture are represented in the HCC. The ability of the FXW-M3

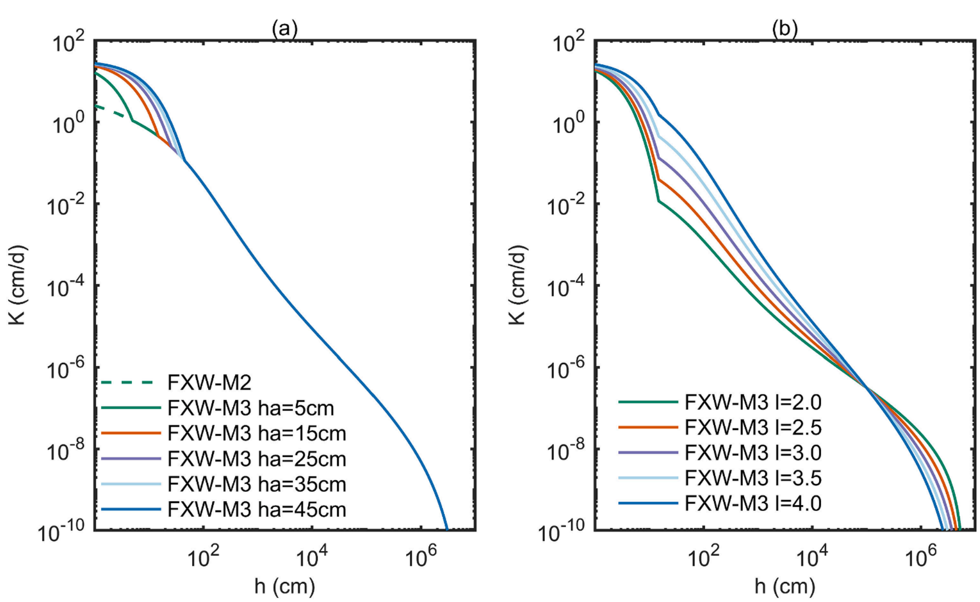

Fig. 1. Illustration of the HCCs of the FXW-M3 model. (a) and (b) present the impact of different values of h_a and l on the HCC prediction, respectively. The other parameters applied are $\alpha = 0.02$ cm^{-1}, $n = 1.01$, $m = 0.66$, $\theta_s = 0.45$ cm^3/cm^3, and $K_s = 31.6$ cm/d for a loam soil.

model to capture the bimodal shape of the HCC is clearly demonstrated in Fig. 1. A higher potential of h_a and a lower value of l represent a sharp decrease in hydraulic conductivity near saturation. However, it should be noted that the SWRC in the FXW-M3 model (Equation 9) is still a unimodal function. As discussed previously in the introduction section, the reason for not applying a bimodal SWRC is to avoid the cost of overparameterization, and the HCC is of more concern than SWRC as water flow in macropores is often assumed to be driven by gravity (Gerke, 2006).

Notably, the saturated matrix conductivity $K(h_a)$ of the FXW-M3 model can be measured or treated as a free-fitted parameter as in other soil hydraulic models developed by Jarvis et al. (1991), Børgesen et al. (2006), and Schaap and van Genuchten (2006) that deal with the influence of soil structure. When there is no measurement, $K(h_a)$ can also be estimated from the known SWRC. As a result, the FXW-M3 model is able to predict the HCC from SWRC with the known matching point of K_s. Furthermore, both capillary and adsorption forces are considered in the FXW-M3 model, allowing a better description of soil hydraulic properties in the low water content range (e.g., Wang et al., 2016; 2018; 2022b).

Additionally, to demonstrate the flexibility of the FXW-M3 model in describing the HCC, we showed the fitted results by treating $K(h_a)$, h_a and l as free-fitting parameters. We refer the model as the FXW-M3-Opt model hereafter.

2.3. The MVGM model developed by Ippisch et al. (2006)

Compared to the original VGM model, the MVGM overcomes the unrealistic decrease of conductivity near saturation for especially fine-textured soils (Ippisch et al., 2006). When the effects of soil structure are important near saturation, the MVGM model tends to overestimate the hydraulic conductivity near saturation (Schaap et al., 2001). The SWRC of the MVGM model is described as

$$S_c(h) = \frac{\theta - \theta_r}{\theta_s - \theta_r} = \begin{cases} \dfrac{\Gamma_{VG}(h)}{\Gamma_{VG}(h_s)} & h < h_s \\ 1 & h \geqslant h_s \end{cases} \tag{10}$$

with $\Gamma_{VG}(h)$ being:

$$\Gamma_{VG}(h) = (1 + |\alpha_{VG} h|^{n_{VG}})^{-m_{VG}} \tag{11}$$

where S_c is the effective water saturation degree, θ_r is the residual water content, and α_{VG} (L^{-1}), n_{VG}, $m_{VG} = 1\text{–}1/n_{VG}$, are fitting parameters. h_s is set to -1 cm following Vogel et al. (2000).

The HCC of the MVGM model is written as

$$K(h) = \begin{cases} K_s S_c^{l_{VG}} \left[\dfrac{1 - \left(1 - (S_c \Gamma_{VG}(h_s))^{1/m_{VG}}\right)^{1-1/n_{VG}}}{1 - \left(1 - \Gamma_{VG}(h_s)^{1/m_{VG}}\right)^{1-1/n_{VG}}} \right]^2 & h < h_s \\ K_s & h \geqslant h_s \end{cases} \tag{12}$$

where l_{VG} is set as 0.5.

3. Materials and methods

3.1. Datasets

The applied datasets were selected from the UNsaturated SOil hydraulic database (UNSODA) (Nemes et al., 2001). Following Zhang et al. (2022), to ensure that the measured data contains sufficient information near saturation, the measured $K(h)$ dataset should include K_s and two more data points at matric potentials higher than -20 cm. However, when it comes to $\theta(h)$, only five data pairs have measured θ_s when the requirement of $K(h)$ is met. Therefore, in order to have a larger amount of data, we specified the measured $\theta(h)$ should have at least two data pairs at matric potentials higher than -10 cm. Meanwhile, both $\theta(h)$ and $K(h)$ should have at least five measured data. As a result, a total of 52 soil samples were selected. Among them, there were seven samples of sand soils, ten samples of sandy loam soils, 11 samples of loam soils and 14 samples of silt loam soils. For loamy sand, silt, silty clay and clay soils, only a limited data size of one, two, one, three respectively are included.

Table 1
The optimized and fixed parameters of different model settings. The number in the bracket demonstrates the lower and upper boundary of the optimized parameters.

Model	SWRC	HCC	Optimized Parameters	Fixed Parameters
MVGM	Eq. (10)	Eq. (12)	α_{VG}(0.001,0.25),1/cm n_{VG} (1.01,10.00) θr(0.01,0.25) θ_s(0.24,0.65)	$h_s = -1$ cm $l_{VG} = 0.5$ K_s
FXW-M2	Eq. (1)	Eq. (3)	α(0.001,0.10),1/cm n(1.01,10.00) m(0.01,1.5) θ_s(0.24,0.65)	$h_s = -1$ cm $l = 3.5$ $b(h_m) = 2.693 \times 10^{-6}$ cm d^{-1}
FXW-M3	Eq. (9)	Eq. (8)	α(0.001,0.10),1/cm n(1.01,10.00) m(0.01,1.5) θ_s(0.24,0.65) h_a* (−0.1,−150),cm	$l = 3.5$ K_s $b(h_m) = 2.693 \times 10^{-6}$ cm d^{-1}
FXW-M3-Opt	Eq. (9)	Eq. (8)	α (0.001,0.10),1/cm n (1.01,10.00) m (0.01,1.5) θ_s (0.24,0.65) h_a (−0.1,−150),cm $K(h_a)$(1.0×10^{-4},1.0×10^2),cm/d l (1.5,8)	K_s

*For the FXW-M3 model, h_a is optimized for each soil type.

3.2. The optimization procedure and the prediction of HCC

We applied four models for comparison, including the MVGM model developed in Ippisch et al. (2006), the FXW-M2 model developed in Wang et al. (2022a), and the FXW-M3 model and the FXW-M3-Opt model proposed in this study. All the optimized and fixed parameters of different model settings are listed in Table 1.

First, the SWRC was fitted with observations to get the parameter sets $p = (\alpha, n, m, \theta_s)$. It should be noted that θ_s in p is only optimized when there is no observation. We apply a Bayesian framework to get the posterior distribution of parameters set p. In general, the posterior distribution of candidate parameter sets p can be expressed as:

$$f(p|\theta_{obs}) = f(\varphi) \prod_{i=1}^{N_\theta} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\theta_{obs}(i) - \theta_{sim}(p,i))^2}{2\sigma^2} \right) \tag{13}$$

where the $f(\varphi)$ is the prior parameter distributions; $f(\varphi|\theta_i)$ is the posterior parameter distributions, N_θ is the number of observations of the water content and θ_{obs} and θ_{sim} are the measured and simulated water contents, respectively, σ is the standard deviation of the model error.

We apply the Particle Evolution Metropolis Sequential Monte Carlo (PEM–SMC) algorithm developed by Zhu et al. (2018) to get the posterior parameter distribution. To approximate the posterior parameter distribution, the PEM-SMC method applies a new sequential Monte Carlo sampler that incorporates the advantages from genetic algorithm, differential evolution and Metropolis-Hasting algorithm. To run the PEM-SMC method, two critical parameters need to be set, including the number of particles and the number of evolutions. In this study, the number of particles was set to 400 and the number of evolutions was set to 300. We refer the readers to Zhu et al. (2018) for details.

When the parameter vector p of the SWRC was determined, the HCC can be predicted by Equations (3) and (14) for the FXW-M2 and the MVGM model, respectively. When it comes to the FXW-M3 model, the prediction of HCC requires the optimal value of h_a. For each soil type, the optimal value for h_a was derived by finding the lowest root-mean-square error of the HCC estimation for h_a varying from −0.1 to −150 cm. When the optimal value of h_a was derived, the HCC was predicted by Equation (8) with the known matching point of K_s.

To evaluate the model performance, the root-mean-square error ($RMSE_{log10(K)}$) and the coefficient of determination (R^2) are calculated. The log-scale value is applied for the hydraulic conductivity.

The $RMSE_{log10(K)}$ is defined as:

$$RMSE_{\log_{10}(K)} = \sqrt{\frac{1}{N_K}\sum_{i=1}^{N_K}\left(\log_{10}(K_i) - \log_{10}(\overline{K}_i)\right)^2} \tag{14}$$

R^2 is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^{N_K}\left(\log_{10}(K_i) - \log_{10}(\overline{K}_i)\right)^2}{\sum_{i=1}^{N_K}\left(\log_{10}(K_i) - \widehat{\log_{10}}(\overline{K})\right)^2} \tag{15}$$

where $\widehat{\log_{10}}(\overline{K})$ is the mean value of $\log_{10}(K_i)$.

4. Results

4.1. The optimal values of h_a for different soil types

Fig. 2 shows the impact of different values of h_a on the performance of the HCC estimation with the FXW-M3 model. A non-zero h_a does improve the estimation of HCC, represented by a higher value of R^2 and a lower value of $RMSE_{log10(K)}$. The improvement is more pronounced for sand, sandy loam, loam and silt loam soils. When it comes to clay soils, only slightly improvement is achieved.

The optimal value of h_a varies for the different soil types of the validated 52 soil samples. For sand and loam soils, the $RMSE_{log10(K)}$ decreases and the R^2 increases sharply for h_a varying from −0.1 to about −50 cm and then changes smoothly for more negative matric potentials, yielding an optimized value of h_a close to the lower boundary of −150 cm. The optimal value of h_a is about −27 cm and −8 cm for sandy loam and silt loam soils, respectively. When it comes to clay soils, h_a has a much higher optimized value of about −4 cm. However, it should be kept in mind that the clay soil types have only three soil samples. For all the 52 soil samples, the optimal value of h_a is about −16 cm.

4.2. The overall performance in predicting the HCC

The optimal value of h_a for each soil type is applied in estimating the HCC. The model performance for the evaluated 52 soil samples and for the five main soil types are summarized in Table 2. Among all models, the MVGM model shows the poorest performance, with the highest $RMSE_{log10(K)}$ value of 1.46 cm d^{-1} and the lowest R^2 value of 0.69 when evaluating with the 52 soil samples. The FXW-M2 model substantially improves the prediction of HCC, with a much lower $RMSE_{log10(K)}$ of 0.90 cm d^{-1} and a higher R^2 of 0.86. When taking into account the effects of soil structure, the FXW-M3 model further improves the prediction of the HCC, reducing the $RMSE_{log10(K)}$ from 0.90 cm d^{-1} (the FXW-M2 model) to 0.78 cm d^{-1} and increasing the R^2 slightly from 0.86 to 0.88. When treating K (h_a), h_a and l as free fitting parameters (the FXW-M3-Opt model), a reported $RMSE_{log10(K)}$ of only 0.23 cm d^{-1} remains, and a much higher R^2 of 0.99 was achieved.

For the five main soil types, the MVGM model generally reports the highest values of $RMSE_{log10(K)}$ except for loam soils, where the FXW-M2 model has the highest value. The FXW-M3 model outperforms the MVGM and the FXW-M2 models for all five soil types, reporting the lowest values of $RMSE_{log10(K)}$ and the highest values of R^2. When treating $K(h_a)$, h_a and l as free-fitting parameters, the FXW-M3-Opt model achieves a substantial improvement for all soil types, with reported $RMSE_{log10(K)}$ ranging from 0.20 cm d^{-1} (loam and silt loam) to 0.36 cm d^{-1} (clay) and reported R^2 all higher than 0.96.

4.3. Predicting the HCC for individual soil samples

4.3.1. Sand soils

Seven soil samples belong to sand soil. For the SWRC, Fig. 3 shows that all models are in excellent agreement with observations for all the seven samples. Due to the difference in model structure, the MVGM model approaches the residual water content while the FXW-M2 and FXW-M3 models show a decrease trend of water content in very dry conditions. However, no observation of water retention data is available in very dry range, hindering the evaluation of the model performance. The observed SWRC generally does not exhibit an obviously bimodal shape for sand soils.

When it comes to HCC prediction, the MVGM model underestimates the conductivity of all soils for matric potentials less than about −1000 cm except for soil 4001, which has no observations in this matric potential range (Fig. 3). The MVGM model overestimates the hydraulic

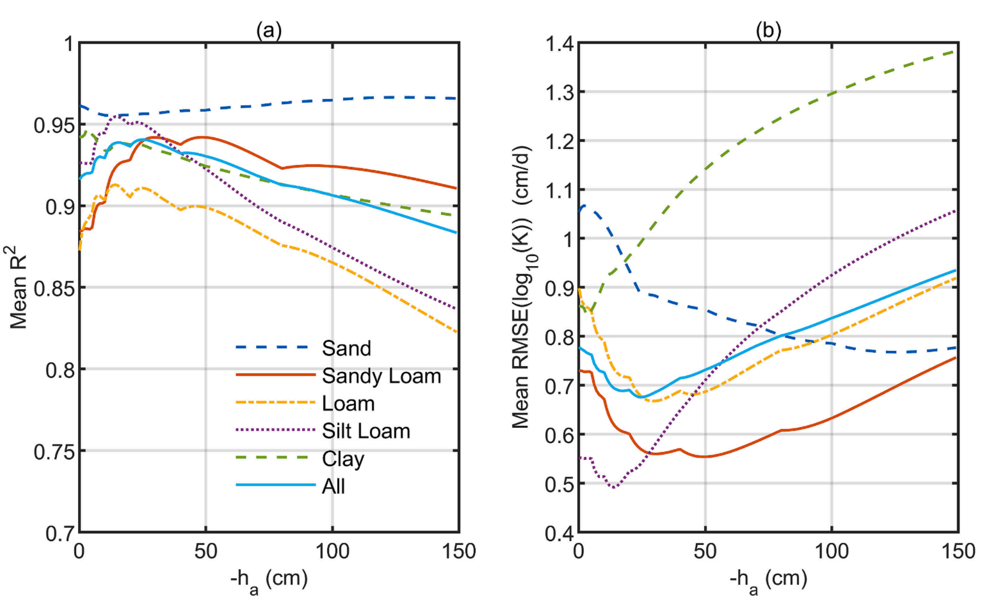

Fig. 2. The prediction of HCC of the FXW-M3 model on different values of h_a for different soil types (a) R^2. (b) $RMSE_{log10(K)}$.

Table 2
Statistical values of different models for each soil type. The number in the bracket is the number of soil samples.

Soil Type		$RMSE_{log10(K)}$ (cm d^{-1})				R^2			
	ha	MVGM	FXW-M2	FXW-M3	FXW-M3-Opt	MVGM	FXW-M2	FXW-M3	FXW-M3-Opt
Sand (7)	149	1.80	0.96	0.79	0.25	0.89	0.87	0.92	0.99
Sandy loam (10)	27	0.95	0.80	0.64	0.21	0.81	0.87	0.92	0.99
Loam (11)	132	0.88	1.07	0.83	0.20	0.89	0.86	0.90	0.99
Silt loam (14)	8	1.37	0.68	0.54	0.20	0.70	0.85	0.92	0.98
Clay (3)	4	2.06	0.89	0.68	0.36	0.68	0.84	0.94	0.96
All (52)	16	1.46	0.90	0.78	0.23	0.69	0.86	0.88	0.99

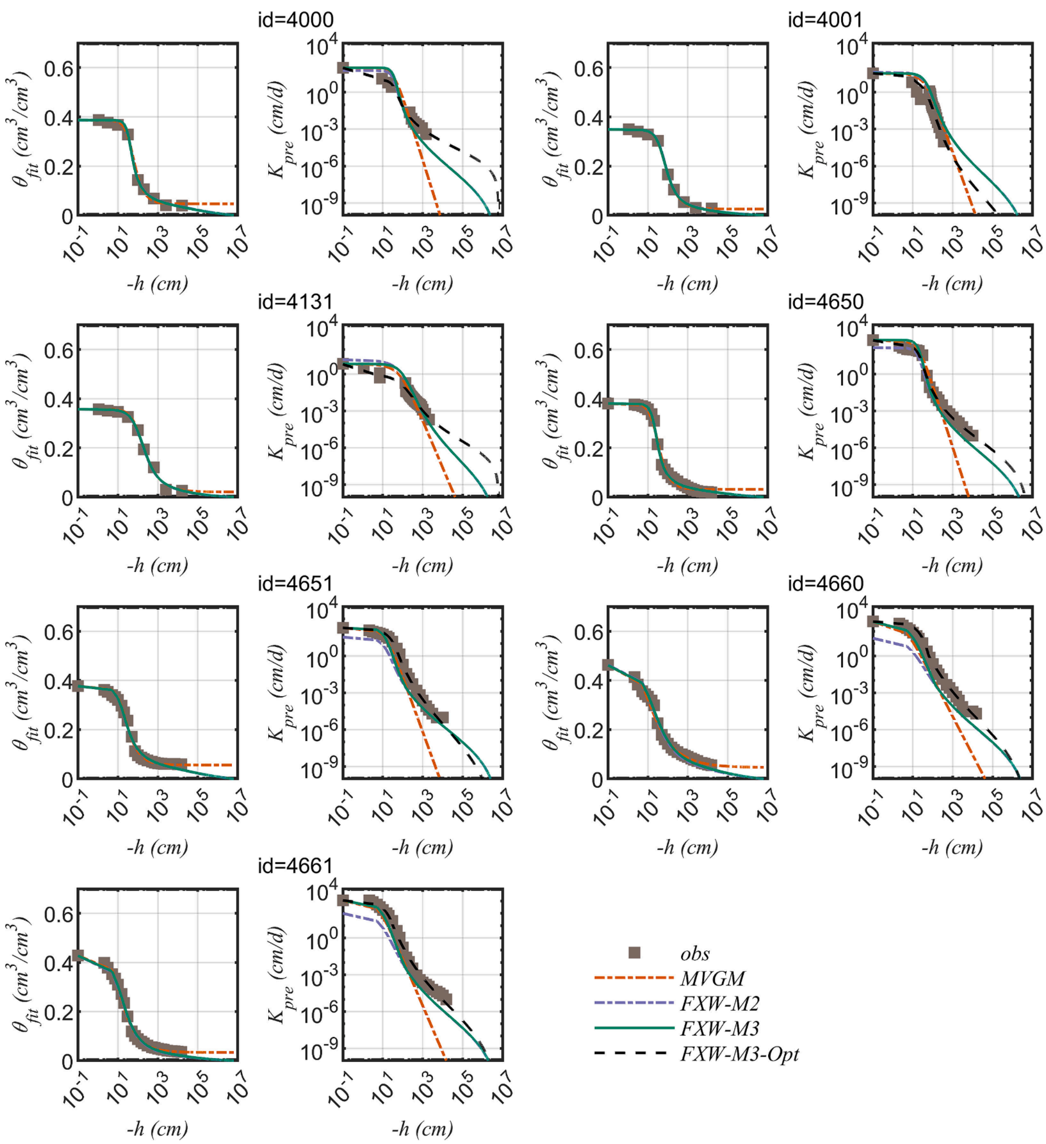

Fig. 3. Predictions of the hydraulic conductivity with the MVGM, FXW-M2, FXW-M3 and the FXW-M3-Opt models for sand soils. For each soil sample (represented by different id in the UNSODA database), the figure in the left is the fitted SWRC while the figure in the right is the predicted HCC.

conductivity near saturation for soil 4131. As a result, the MVGM model reported the highest $RMSE_{log10(K)}$ value of 1.80 cm d^{-1} for sand soils (Table 2). Compared to the MVGM model, both the FXW-M2 and the FXW-M3 models improve the prediction of conductivity in dry conditions. The FXW-M2 model reported a much lower value of $RMSE_{log10(K)}$ of 0.96 cm d^{-1} and generally showed a close agreement with observations for soils 4000, 4001 and 4650. However, mainly for soils 4651, 4660 and 4661, the FXW-M2 model underestimates the conductivity for almost all the measured matric potential range. As a result, a much negative value of −149 cm was derived for the optimal h_a for the FXW-M3 model. The FXW-M3 model notably improves the prediction of HCC near saturation, with the value of $RMSE_{log10(K)}$ reducing from 0.96 cm d^{-1} of the FXW-M2 model to 0.79 cm d^{-1}. However, according to the criteria set for the importance role of soil structure, i.e., the underestimation by the FXW-M2 model and the overestimation by the MVGM model of near-saturated hydraulic conductivity, the effects of soil structure on the seven sandy soils were concluded as not significant. When treating $K(h_a)$, h_a and l as free fitting parameters, a $RMSE_{log10(K)}$ of

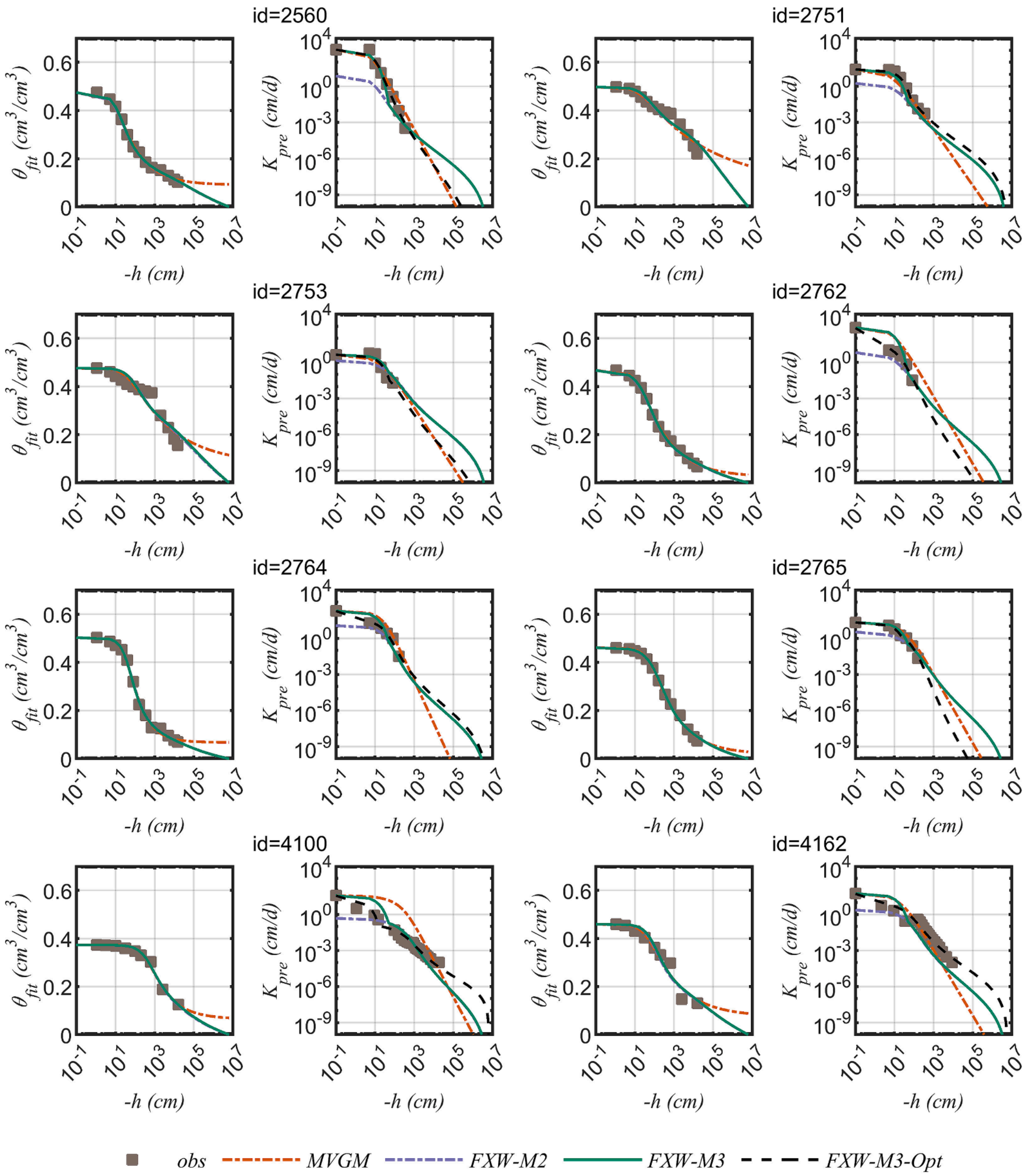

Fig. 4. Predictions of the hydraulic conductivity with the MVGM, FXW-M2, FXW-M3 and the FXW-M3-Opt models for sandy loam soils. For each soil sample (represented by different id in the UNSODA database), the figure in the left is the fitted SWRC while the figure in the right is the predicted HCC.

only 0.25 cm d^{-1} remains for the seven soil samples.

4.3.2. Sandy loam soils

For the ten sandy loam soil samples, all models are generally in close agreements with observations in fitting the SWRC (Fig. 4 and Figure S1). The deviation happens for soils 2751, and 2753 (Fig. 4), where the retention data shows an obvious bimodal shape. The structure bimodality, which may be due to the effects of a bimodal soil particle size distribution (Dexter et al., 2008), occurs generally close to the matric potential of about −1000 cm. However, the measured hydraulic conductivity does not reach this matric potential range, hindering the evaluation of the impact of bimodal SWRC on the estimation of HCC. The FXW-M2 model underestimates the hydraulic conductivity near saturation for all ten sandy loam soils, with a reported $RMSE_{log10(K)}$ of 0.80 cm d^{-1} and R^2 of 0.87. Meanwhile, the overestimation of the MVGM model in predicting the HCC is noticed for soils 2762, 2764, 4100 in Fig. 4 and 4162, 4171 in Figure S1, indicating clearly the effects of soil structure. The reported $RMSE_{log10(K)}$ is 0.95 cm d^{-1} and R^2 is 0.81 for the MVGM model. Compared to the FXW-M2 model, the FXW-M3 model improves the prediction of conductivity for all ten sandy loam soils, with reported $RMSE_{log10(K)}$ reduces from 0.80 of the FXW-M2 model to 0.64 cm d^{-1} and R^2 increases from 0.87 to 0.92. However, for soils 2762, 2764, 4110, 4162 and 4172, the FXW-M3 model overestimates the hydraulic conductivity near saturation, indicating a higher h_a value than the optimized −27 cm for these five soil samples. The model performance can be substantially improved when treating $K(h_a)$, h_a and l as free fitting parameters, with reported $RMSE_{log10(K)}$ of 0.21 cm d^{-1} and R^2 of 0.99 remain (Table 2).

4.3.3. Loam soils

For the 11 loam soils, a slight deviation in fitting the SWRC is noticed for soils 2601, 2602, 2611, 2750 and 2752, where a bimodal SWRC is shown in the mid-water content range (Fig. 5). The MVGM model is generally in close agreement with observations when predicting the HCC for seven out of 11 loam soil samples (Fig. 5). For soils 2601, 2602, 4102 and 4111, the MVGM model overestimates the hydraulic conductivity near saturation. The reported $RMSE_{log10(K)}$ is 0.88 cm d^{-1} and R^2 is 0.89. The FXW-M2 model underestimates the hydraulic

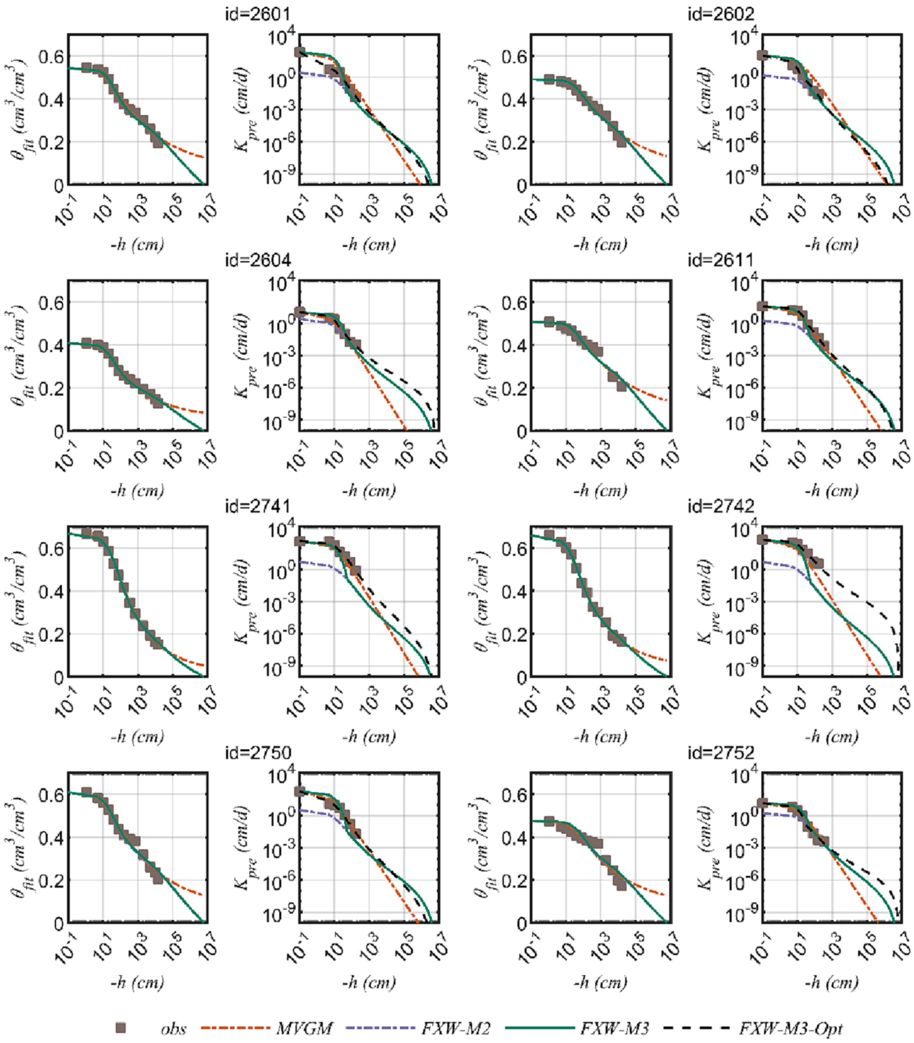

Fig. 5. Predictions of the hydraulic conductivity with the MVGM, FXW-M2, FXW-M3 and the FXW-M3-Opt models for loam soils. For each soil sample (represented by different id in the UNSODA database), the figure in the left is the fitted SWRC while the figure in the right is the predicted HCC.

conductivity near saturation for almost all 11 loam soil samples, reporting a higher $RMSE_{log10(K)}$ of 1.07 cm d^{-1} and a lower R^2 of 0.86 than those reported by the MVGM model (Table 2). The critical matric potential for the FXW-M2 model to underestimate the hydraulic conductivity ranged from about −10 cm for soils 2604, 2752, 4101 and 4102 to negative hundreds of cm for soils 2611, 2741, 2742, 2750. As a result, the optimal h_a of the FXW-M3 model is about −132 cm when evaluating with the 11 loam soils. According to the criteria set for the importance of soil structure, the effects of soil structure are concluded to be important for soils 2601, 2602, 4102 and 4111.

Compared to the FXW-M2 model, the FXW-M3 model improves the prediction near saturation for all 11 loam soil samples (Fig. 5 and Figure S4). The $RMSE_{log10(K)}$ decreases from 1.07 cm d^{-1} of the FXW-M2 model to 0.83 cm d^{-1} of the FXW-M3 model, and the R^2 increases from 0.86 to 0.90 (Table 2). Due to the much negative optimal value of h_a, the FXW-M3 model slightly overestimates the hydraulic conductivity near saturation for soils 2601, 4102 and 4111, indicating the h_a value is in fact different for different soils. The $RMSE_{log10(K)}$ is only 0.20 cm d^{-1} and the R^2 is 0.99 when treating $K(h_a)$, h_a and l as free-fitting parameters.

4.3.4. Silty loam soils

All models are generally in close agreement with observations in fitting the SWRC, except for soils 2612, 2613, 2760 and 2761, where a bimodal SWRC is noticed (Fig. 6). The structure bimodality occurs in the mid-water content range, where no measurements of hydraulic conductivity are available.

When it comes to the HCC, the MVGM model overestimates while the FXW-M2 model underestimates the hydraulic conductivity near saturation for almost all 14 silty loam soils, indicating clearly the effects of soil structure. The reported $RMSE_{log10(K)}$ is 1.37 cm d^{-1} and 0.68 cm d^{-1} and the R^2 is 0.70 and 0.85 for the MVGM and the FXW-M2 model, respectively (Table 2). When accounting for the effects of soil structure, the FXW-M3 model improves the estimation of HCC near saturation in comparison with the FXW-M2 model, reducing the $RMSE_{log10(K)}$ from 0.68 cm d^{-1} of the FXW-M2 model to 0.54 cm d^{-1}, and increasing the R^2 from 0.85 to 0.92. For the 14 silty loam soils, the effects of soil structure are generally important for matric potentials higher than − 10 cm. Treating $K(h_a)$, h_a and l as free fitting parameters substantially reduces the $RMSE_{log10(K)}$ to 0.20 cm d^{-1} and increases the R^2 to 0.98.

4.3.5. Clay soils

The validated clay soils include only three soil samples. All three soils show clearly the effects of soil structure, indicated by the overestimation of the MVGM model and the underestimation of the FXW-M2 model in estimating the near saturated hydraulic conductivity (Fig. 7). The reported $RMSE_{log10(K)}$ is 2.06 cm d^{-1} and 0.89 cm d^{-1}, and the R^2 is 0.68 and 0.84 for the MVGM model and the FXW-M2 model, respectively (Table 2). When applying the FXW-M3 model that accounts for the effects of soil structure, the $RMSE_{log10(K)}$ reduces from 0.89 cm d^{-1} of the FXW-M2 model to 0.68 cm d^{-1} and the R^2 increases from 0.84 of the

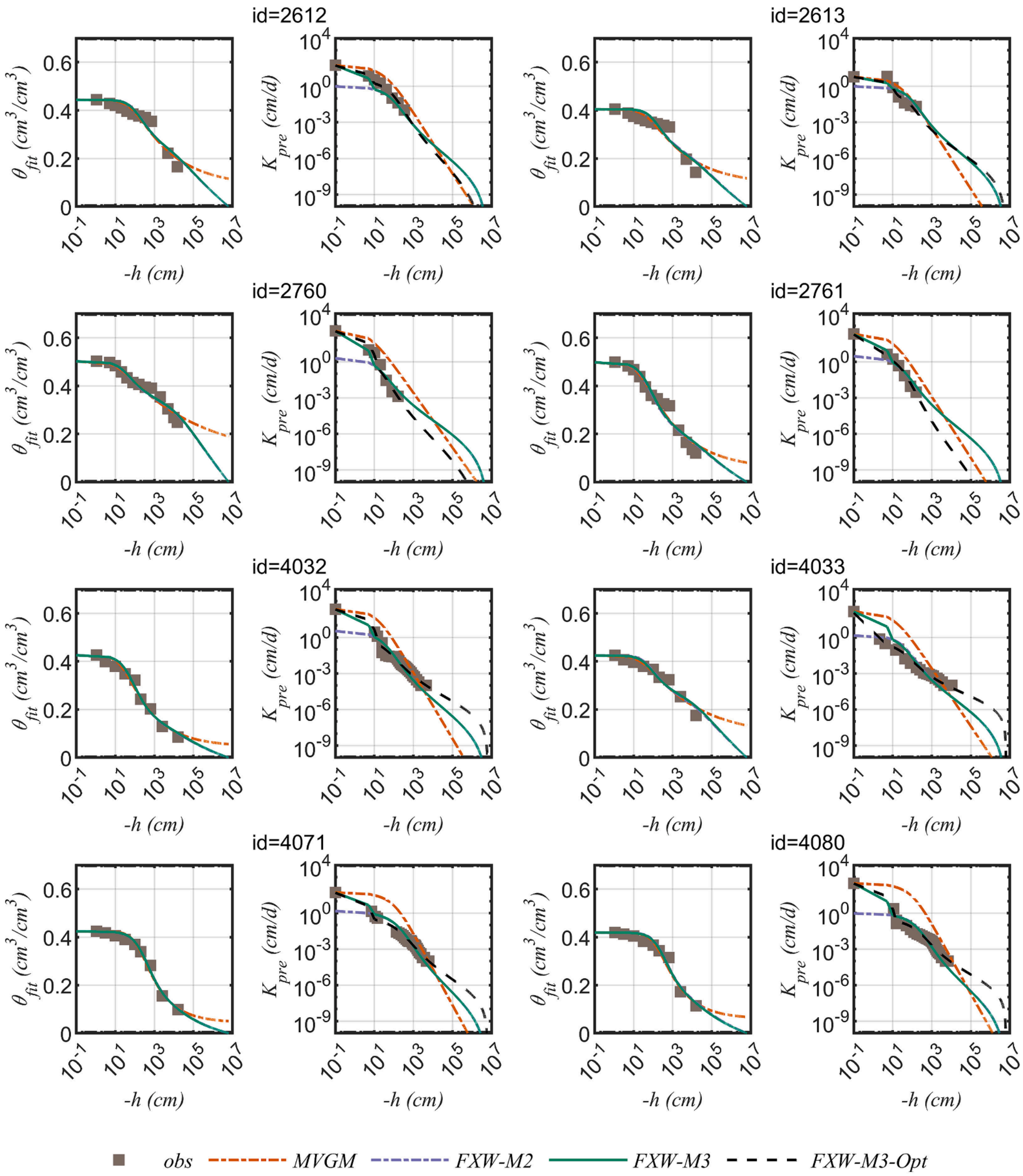

Fig. 6. Predictions of the hydraulic conductivity with the MVGM, FXW-M2, FXW-M3 and the FXW-M3-Opt models for silty loam soils. For each soil sample (represented by different id in the UNSODA database), the figure in the left is the fitted SWRC while the figure in the right is the predicted HCC.

FXW-M2 model to 0.94. For three clay soils evaluated, the optimal h_a has a much higher value of −4 cm than the values of other soil types. The FXW-M3-Opt model that treating $K(h_a)$, h_a and l as free fitted parameters reduces further the $RMSE_{log10(K)}$ to 0.36 cm d^{-1} and increases the R^2 to 0.96.

5. Discussion

5.1. Effects of soil structure on different soil types

For the evaluated 52 soil samples, the MVGM model developed by Ippisch et al. (2006) underestimates the hydraulic conductivity in the dry range for almost all seven sand soils while overestimates the hydraulic conductivity near saturation for almost all 14 silt loam soils, all three clay soils, three out of ten sandy loam soils and three out of 11 loam soils. The underestimation, as discussed in detail in Wang et al. (2018; 2022a; b), is because the MVGM model does not consider the effects of adsorption forces which are dominant in dry conditions. The overestimation near saturation can be attributed to the application of the matching point K_s that is sensitive to the presence of macropores, while the MVGM model only accounts for the impact of soil texture (van Genuchten and Nielsen, 1985). The FXW-M2 model underestimates the hydraulic conductivity near saturation for almost all 52 soil samples. The main reason for the underestimation is that the FXW-M2 model does not address the impact of soil structure (Wang et al., 2022a). Besides, the uncertainty in estimating the parameter $b(h_m)$ may also be the reason of the underestimation of hydraulic conductivity by the FXW-M2 model (Wang et al., 2022a). When accounting for the effects of soil structure near saturation, the proposed FXW-M3 model outperforms the MVGM and the FXW-M2 models for almost all 52 soil samples, represented by a lower value of $RMSE_{log10(K)}$.

However, according to the criteria set, only when the MVGM model overestimates while the FXW-M2 model underestimates the hydraulic conductivity near saturation, we can conclude that the effects of soil structure near saturation are important for a given soil. As a result, for the 52 evaluated soil samples, the effects of soil structure near saturation are concluded to be important for almost all 14 silt loam soils, all three clay soils, three out of ten sandy loam soils and three out of 11 loam

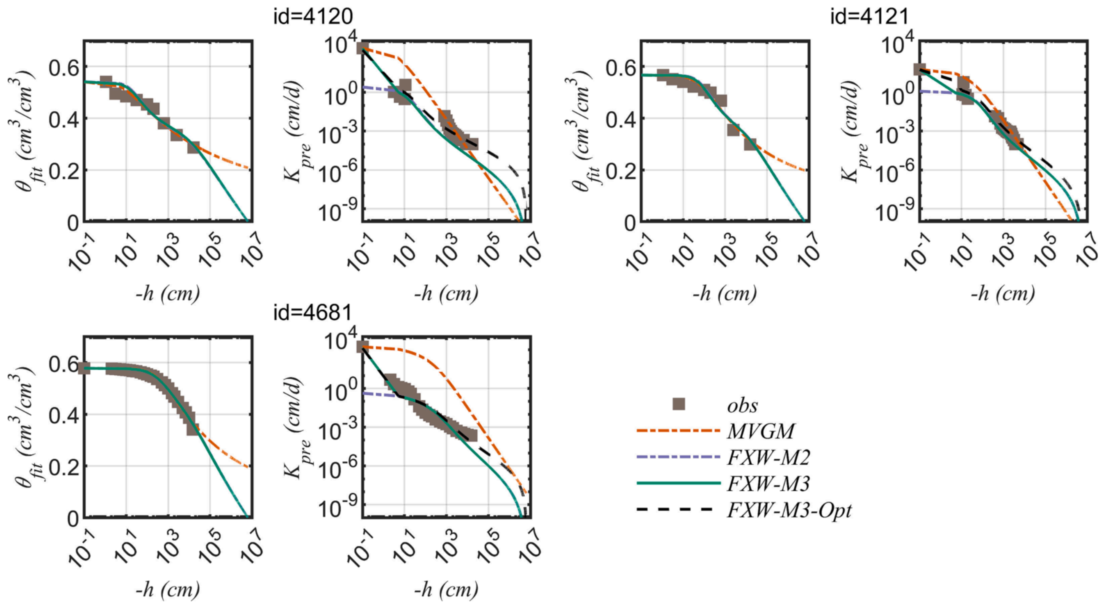

Fig. 7. Predictions of the hydraulic conductivity with the MVGM, FXW-M2, FXW-M3 and the FXW-M3-Opt models for clay soils. For each soil sample (represented by different id in the UNSODA database), the figure in the left is the fitted SWRC while the figure in the right is the predicted HCC.

soils. While for all the seven sandy soils, seven sandy loam soils and eight loam soils, the influence of soil structure near saturation is not significant. For the soil samples where the effects of soil structure near saturation are not significant, the outperformance of the FXW-M3 model to the FXW-M2 model can be attributed to the underestimated parameter $b(h_m)$ or $K(h_a)$.

The possible underestimation of parameter $K(h_a)$ in the FXW-M3 model and the small data size evaluated also explain why the optimal h_a has a much negative value close to −150 cm for the evaluated seven sand soils and 11 loam soils. For three out of seven sand soils (Fig. 3) and four out of 11 loam soils (Fig. 5), the FXW-M3 model considerably underestimated the hydraulic conductivity in the relatively low matric potential range. Due to the small data size evaluated, a much negative value of h_a is thus required to give the lowest $RMSE_{log10(K)}$. According to the definition, h_a means the critical potential separating the influence of soil structure and soil texture. Therefore, it is expected that h_a has a much higher value. For example, Scotter (1978) suggested the critical matric potential h_a is about −15 cm to account for macropore, Børgesen et al. (2006) suggested a value of −4 cm, whereas Jarvis (2007) suggested the value as −10 to −6 cm. Therefore, the optimal h_a value derived here for different soil type should not be interpreted strictly as an effect of soil structure. When evaluating with all the 52 soil samples, the optimal h_a value is about −16 cm, much close to the reported value in the literature. In fact, the fitting results of the FXW-M3-Opt model suggested although h_a generally varies in a narrow range, the value can be different for different soil sample.

Besides the effects of soil structure near saturation, the effects of soil structure in the mid-water content range were found for two sandy loam soils, three loam soils and five silt loam soils, represented by a bimodal shape SWRC (Figs. 4-6). Dexter et al. (2008) also concluded that the bimodal SWRC are normal in sandy loam and loamy sand soils while it is less common in pure sand soils. This structural bimodality that occurs in the mid-water content range could be attributed to the impacts of a bimodal particle size distribution (Dexter et al., 2008).

5.2. Model limitations

When the FXW-M3 model is generally in good agreement with observations, it underestimates the hydraulic conductivity for about seven out of 52 soil samples. As discussed previously, this underestimation is attributed to the uncertainty in estimation of $b(h_m)$ or $K(h_a)$. Wang et al. (2022a) presented a detail discussion on the uncertainty in estimating $b(h_m)$, showing many factors such as the applied value of the Hamaker constant, the ionic concentration (Tokunaga, 2011) and the possible impact of capillary forces in the dry range all have an impact on the estimation of $b(h_m)$. Therefore, if the saturated matrix hydraulic conductivity $K(h_a)$ is measured, it is preferred to use the measured value directly as a matching point.

Another limitation of the FXW-M3 model is that it only accounts for the effects of soil structure near saturation. For two sandy loam soils, three loam soils and five silt loam soils, the SWRC shows a bimodal shape. This structural bimodality that occurs in the mid-water content range cannot be captured by the proposed FXW-M3 model. However, it is hard to evaluate the impact of this bimodal SWRC on the estimated hydraulic conductivity because the measured hydraulic conductivity generally does not reach this potential range. To fully describe the influence of soil structure that occurs both near saturation (due to the presence of macropore) and far from saturation (due to a bimodal particle size distribution), the bimodal or multimodal SWRC should also be applied in addition to modifying the HCC. However, as discussed in the introduction section, the application of the bimodal or multimodal SWRC introduces too many parameters and requires a detailed measurement of water retention data (e.g., Othmer et al., 1991, Ross & Smettem, 1993; Durner, 1994). This complex bimodal soil hydraulic model may be useful in profile applications but is rarely applied in large-scale applications (Fatichi et al., 2020).

5.3. Data limitations

In this study, 52 soil samples from the UNSODA database were applied for model evaluation. However, only 18 soil samples of them have more than three measured data pairs and one been still K_s for matric potentials higher than −10 cm, where soil structure is known to exert a strong influence on hydraulic conductivity (Jarvis, 2007). Meanwhile, only five soil samples have measured value of θ_s. The limited data size and the limited information close to saturation thus hinder an accurate assessment of the model performance. For the ten soil samples that exhibited bimodal SWRC, the measured HCC generally covered only a relatively high range of matrix potentials. As a result, it is also difficult to assess the impact of bimodal SWRC on HCC prediction.

6. Concluding remarks

In this study, we developed a novel model for the prediction of HCC by considering the effects of soil structure near saturation. The introduced saturated matrix hydraulic conductivity in the termed FXW-M3 model can either be measured using tension infiltrometers (Jorda et al., 2015) or be estimated from the SWRC. In addition, the new FXW-M3 model takes into account both the capillary and adsorption forces, providing a much better description of soil hydraulic properties in the low water content range compared to the commonly applied capillary-based models (Wang et al., 2016). Therefore, the FXW-M3 model accounts for both the impact of soil structure near saturation and soil texture and is able to describe the soil hydraulic properties from saturation to oven dryness. Compared to the well-known van Genuchten (1980)-Mualem (1976) model, the FXW-M3 model introduces no additional free parameters. It is worth noting that in the FXW-M3 model, only the effects of soil structure on HCC are considered, while the effects on SWRC are considered to be of less concern. Testing with 52 soil samples suggested the FXW-M3 model, with $K(h_a)$ estimated from the SWRC, outperformed the FXW-M2 model developed in Wang et al. (2022a) and was generally in close agreement with observations.

Therefore, this new FXW-M3 model provides an easy and practical way to incorporate the influence of soil structure near saturation in water and solute transport simulations. However, due to the uncertainty in estimating $K(h_a)$ from the SWRC, the FXW-M3 model may underestimate the hydraulic conductivity for some soil samples, indicating it is preferring to use the measured $K(h_a)$ as matching point when available. In addition, the FXW-M3 model also cannot capture the structural bimodality that occurs in the mid-water content range, which is found in a few soil samples. Finally, only 52 soil samples with limited measurements near saturation were applied for model evaluation. Therefore, a large data size with high quality measurements of soil hydraulic properties are further needed to better assess the performance of the proposed model and the optimal value of h_a.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The link to the data is shown in the Acknowledgements section

Acknowledgments

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jhydrol.2023.129330.

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